

AI Acceptance Across Service Platforms

From Discussion to Design — Integrating Reddit and Trustpilot

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1 The idea and research question

Online services increasingly put **AI and AI agents** between the user and the thing they came for: chatbots answer support tickets, algorithms screen and price, automated systems make decisions. We study **how people accept (or reject) that AI** — and, crucially, how acceptance changes with the *role* AI plays. We compare three service contexts:

- **AI-native services** — the product *is* an AI agent (ChatGPT, Claude, Replika, Gemini, Perplexity);
- **Peer-to-peer rentals** — AI is a *peripheral* automation layer on a human marketplace (Turo, Airbnb);
- **Customer service** — AI runs the *support* layer (chatbots replacing agents).

The middle context is the assignment’s heart — a **peer-to-peer sharing economy** for high-value professional gear: cameras and drones (Fat Llama), photography lenses (Lensrentals), film and event kit (KitSplit), vehicles (Turo, Getaround) and RVs (Outdoorsy, RVshare) — while the AI-native and customer-service contexts act as comparison lenses that reveal how acceptance shifts as AI moves from *being* the product, to a marketplace add-on, to the support desk.

Research question. How do users accept and experience AI agents across different service platforms — and what drives or erodes trust as AI moves from being the product, to a marketplace add-on, to the support desk?

We use two complementary corpora. **Reddit is the discussion baseline** — open deliberation where users describe AI experiences in their own words. **Trustpilot is the verification layer** — reviews carrying a **1–5 star rating** that acts as a ground-truth acceptance label. Where the open discussion and the labelled reviews agree we gain confidence; where they diverge we learn something real.

2 Data in brief

Reddit (reddit_baseline.csv). 4,269 comments across 12 subreddits (Nov 2023 – May 2026), pulled from the **Arctic Shift** research archive (01_fetch_reddit_arcticshift.R) because Reddit’s 2026 anti-bot wall blocks live scraping. AI-native subreddits are pulled in full; rental and support subreddits by AI/automation keyword search. Each comment is tagged with a **platform_category**: ai_service 2027, rental 1715, customer_service 527. Top subreddits: r/turo (747), r/AirBnB (578), r/CustomerService (527), r/airbnb_hosts (390), r/artificial (270), r/replika (270).

Trustpilot (trustpilot_flagged.csv). 3,912 reviews from seven peer-to-peer rental platforms (Apify actor casper11515/trustpilot-reviews-scraper, merged by 02_merge_trustpilot.R), each with a

1–5 star label. `06_ai_acceptance.R` flags the **60 reviews that describe an AI / automation experience** (bots, algorithms, automated decisions, “no human”).

3 Method

The pipeline runs in five steps. **Steps 1–4** (`steps 1 to 4.R`) operate on the Reddit baseline: (1) **preprocessing** — clean, lowercase, stop-word removal, lemmatisation, and a two-class split (positive vs negative) from lexicon sentiment; (2) **exploratory** — top-10 words per class, commonality and comparison word clouds (1- and 2-gram); (3) **structural & sentiment** — bigram co-occurrence network and an NRC emotion comparison; (4) **modelling** — LDA topics per class (1- and 2-gram) and GloVe embeddings for the neighbours of *trust*, *bot*, *host*.

Step 5 (`05_integration.R`, this report) is the synthesis layer. It brings in Trustpilot and fuses the two sources three ways: a **convergent-validity test** of the Reddit method against real stars, a **shared theme lexicon** applied to both corpora, and the **AI-role gradient**. For portability it uses the bundled Bing lexicon (sign-of-sum, the same two-class logic as the AFINN classing in steps 1–4) and base-R I/O, so it runs with no downloads.

4 Findings: integrating the two sources

The brief asks how the **word networks, sentiments, and topics overlap to tell a cohesive story**. They do, and Trustpilot lets us prove it rather than assert it. The negative-class structures that steps 1–4 surface on Reddit — the co-occurrence hubs around *bot / human / support*, the negative-leaning NRC emotions, and the LDA topics about automation and unresolved problems — reappear as the themes that drive **low star ratings** on Trustpilot. Three results carry the story.

4.1 The Reddit method is valid: lexicon sentiment tracks real stars

Reddit has no ground-truth label, so steps 1–4 *infer* the positive/negative class from a sentiment lexicon. Is that inference trustworthy? Trustpilot is the only source with **both** a lexicon score and a real star rating, so it is where we can test it. Applying the identical lexicon class to Trustpilot text and comparing it to the star-based class gives **93.4% agreement** (Cohen’s $\kappa = 0.76$, $n = 3,534$; Figure 1) — *substantial* agreement by the usual benchmark. The lexicon recovers 76% of genuinely negative reviews. This validates using lexicon sentiment as the class on Reddit, and it is the methodological bridge that lets us read the two corpora together.

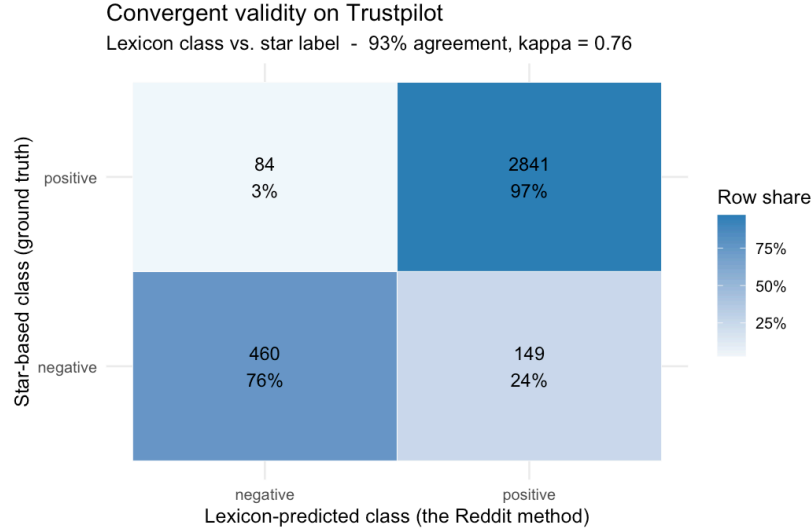


Figure 1: Lexicon-predicted class vs. the 1-5 star ground truth on Trustpilot. High agreement validates the sentiment-based classes used on Reddit.

4.2 What the two sources agree on: the drivers of frustration

We tag every comment and review with the same six AI-acceptance themes and compute, per source, a **lift** = how much more often a theme appears in negative than in positive text (Figure 2). Four themes exceed 1 in *both* sources — they are **cross-validated frustration drivers**, robust across two structurally different corpora:

Table 1: Negativity lift by theme and source (>1 = drives the negative class).

Theme	Reddit lift	Trustpilot lift	Cross-validated
Automation vs. human	1.05	8.07	yes
Accuracy & quality	1.77	6.85	yes
Pricing & fees	0.71	5.42	rental-only
Trust & safety	0.86	4.94	rental-only
Support & resolution	1.17	1.77	yes
Anthropomorphism / companionship	1.49	1.45	yes

The cleanest is **accuracy & quality** (errors, “doesn’t work”, hallucination: 1.8x on Reddit, 6.9x on Trustpilot) followed by **automation-vs-human** (the “no human / bot” signature: pervasive on Reddit, an 8x marker of negative reviews on Trustpilot) and **support & resolution**. Two themes — **pricing & fees** and **trust & safety** — spike only on Trustpilot: they are *rental-specific* transaction frustrations, not complaints about the AI itself, which is exactly why the two-source design matters (Reddit alone would have missed that distinction).

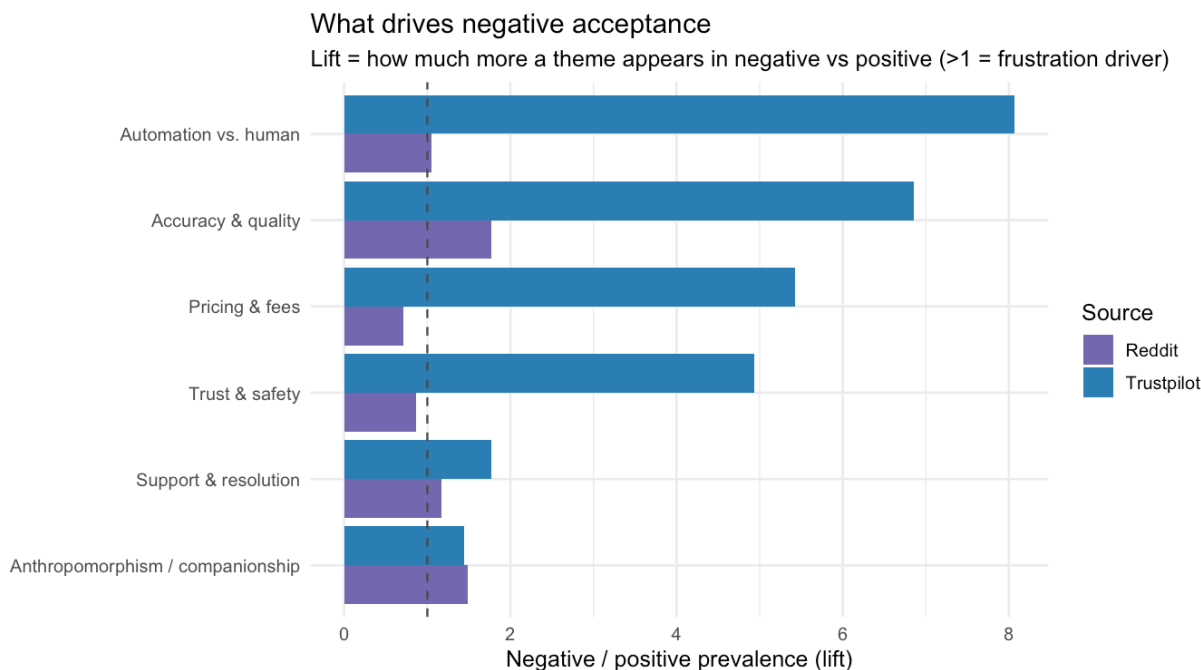


Figure 2: Negativity lift by theme and source: four themes clear the dashed line (lift = 1) in both Reddit and Trustpilot.

4.3 The AI-role gradient: acceptance depends on what AI *is*

Acceptance is not a fixed attitude to “AI” — it tracks the **role** AI plays. On Reddit, the share of positive comments is highest where users *chose* an AI product and lowest where AI is bolted onto a rental transaction:

Table 2: Reddit: positive-class share by AI role.

Context (Reddit)	Comments	% positive
AI-native (the product)	1266	59%
Customer service (the support desk)	397	59%
Rental (a marketplace add-on)	1290	51%

Trustpilot sharpens the point with the **AI / automation penalty**: rental reviews that mention AI or automation average **2.32 stars** against **4.21** for the rest — a **1.89-star** gap (Figure 3). Same technology, opposite reception: welcomed when it *is* the chosen service, penalised when it stands between a user and a human transaction.

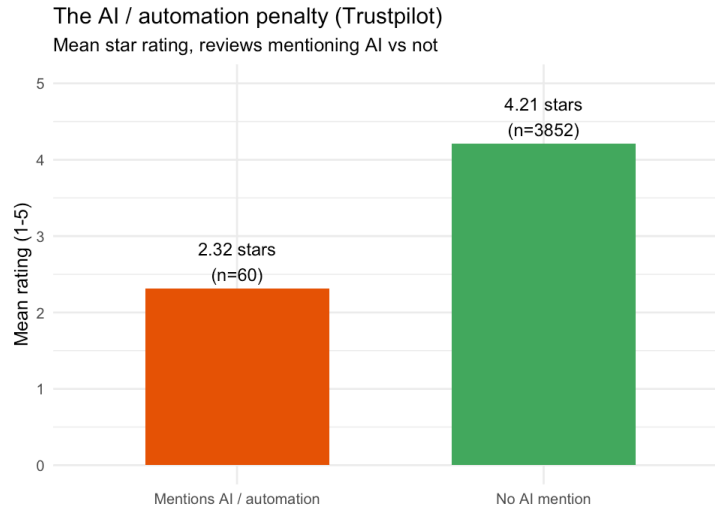


Figure 3: The AI/automation penalty on Trustpilot: mentioning AI coincides with a 1.9-star drop.

5 Strategic insights: designing AI agents for acceptance

Read together, the three findings convert into actionable design guidance. Each card below pairs a cross-source finding with a design principle and the trust/adoption theory it rests on.

Automation vs. human (*Cross-source*). Automation/‘no human’ language is 8.1x more common in negative Trustpilot reviews and pervasive in Reddit AI talk (lift 1.05) – the signature of forced automation.

- *Design principle*: Make the human hand-off one click away; never trap users in a bot loop. Label AI as AI and state its limits up front.
- *Anchored in*: Algorithm aversion (Dietvorst et al. 2015); TAM perceived usefulness (Davis 1989)

Accuracy & quality (*Cross-source*). Errors / ‘doesn’t work’ / hallucination is the cleanest driver: 1.8x (Reddit) and 6.9x (Trustpilot) more common in negative text.

- *Design principle*: Show confidence and sources; let users correct the agent; fail loudly, not silently.
- *Anchored in*: Ability & integrity trust (Mayer et al. 1995)

Support & resolution (*Cross-source*). Slow / unresolved support drives the negative class in both sources (lift 1.2 Reddit / 1.8 Trustpilot); resolution speed beats politeness.

- *Design principle*: Optimise the agent for first-contact resolution and a transparent wait/status, not chat length.
- *Anchored in*: UTAUT effort expectancy & facilitating conditions (Venkatesh et al. 2003)

Anthropomorphism / companionship (*Cross-source (double-edged)*). Persona/companionship language is emotionally loaded and tilts slightly negative in both sources (lift ~1.5) – strong delight when it works, strong frustration when the persona disappoints.

- *Design principle*: Match persona warmth to role (personable for AI-native companions, low-key for transactional add-ons) and keep the persona stable – abrupt changes break trust.
- *Anchored in*: CASA paradigm (Nass & Moon 2000); expectation-confirmation (Bhattacharjee 2001)

Pricing & fees (*Rental-specific*). Opaque fees / refunds dominate negative rental reviews (5.4x on Trustpilot) but are NOT an AI-native frustration (lift 0.7 on Reddit).

- *Design principle*: Surface all-in price and refund rules before commitment; let the agent proactively explain charges.
- *Anchored in*: Benevolence trust (Mayer et al. 1995); price-fairness

Trust & safety (*Rental-specific*). Trust / scam / verification language marks negative rental reviews (4.9x on Trustpilot), less so on Reddit (0.9x).

- *Design principle*: Foreground verification, security and data-use cues; let the agent cite its safeguards.
- *Anchored in*: Initial trust formation (McKnight et al. 2002)

The headline recommendation. Treat the human hand-off as a feature, not a failure: the single most consistent signal across both sources is that *forced* automation — being trapped with a bot and unable to reach a person — destroys acceptance, while *chosen* automation earns it. AI agents should be **transparent** (labelled as AI, honest about limits), **escapable** (one-click human hand-off), and **role-matched** (a personable companion for AI-native products; an efficient, near-invisible helper for transactional add-ons). This mirrors *task-dependent* algorithm aversion (Castelo et al. 2019): users accept AI for objective, mechanical tasks and resist it for tasks they see as needing human judgement.

6 Progress against the assignment brief

Each task is tagged **Done**, **In progress**, or **To do**.

- **Dataset & preprocessing** — **Done**. Two self-scraped corpora, no X/Twitter: Reddit (4,269 comments, 3 contexts) and Trustpilot (3,912 star-labelled reviews + AI flag). Cleaning, lemmatisation, tokenisation and the two-class split are implemented in **steps 1 to 4.R**.
- **Exploratory (word frequency, clouds)** — **Done**. Top-10 words per class (bar charts) and commonality / comparison clouds, 1- and 2-gram (**steps 1 to 4.R**, step 2).
- **Structural & sentiment (networks, sentiment)** — **Done**. Bigram co-occurrence network and NRC emotion comparison across the two classes (step 3).
- **Advanced modelling (LDA, embeddings)** — **Done**. 3–5 LDA topics per class on 1- and 2-gram tokens, plus GloVe neighbours of *trust* / *bot* / *host* (step 4).
- **Integration** — **Done**. Validity test, shared-theme analysis and the AI-role gradient fuse the networks, sentiments and topics into one story and validate it against Trustpilot stars (05_integration.R; Section 4).
- **Strategic insights** — **Done**. Six evidence-backed design cards and a headline recommendation (Section 5).
- **Literature** — **Done**. Methodology and interpretation anchored in the trust/adoption frameworks below (Section 8).
- **To do**. Optional: extend Trustpilot to AI-native platforms (character.ai, replika.com, openai.com) to populate the Trustpilot `ai_service` cell; final presentation polish.

7 Limitations

- Trustpilot skews positive (opt-in bias) — we down-sample for balanced classification; the validity test already uses the natural distribution.
- The Reddit `rental` pull includes a broad `app` query that adds some general app-experience noise; every row keeps a `source_query` column for precise filtering.
- The Trustpilot `ai_service` context is not yet populated (AI-native platforms still to scrape), so the AI-role gradient uses Reddit for the AI-native cell.

- “Trust” is mostly *implicit* (bot/human, automated, algorithm language), measured via themes and sentiment, not keyword counts. English-only.
- The portable Bing lexicon (sign-of-sum) is a coarser instrument than AFINN’s graded scores; the 93.4% star agreement shows it is good enough for classing, but magnitudes should be read as directional.
- Reddit data is archival (Arctic Shift); the live anti-bot wall prevents fresh anonymous scraping.

8 Literature

Our methods and interpretations rest on established work in technology acceptance, trust, and human–AI interaction.

Bhattacharjee, A. (2001). Understanding information systems continuance: An expectation-confirmation model. *MIS Quarterly*, 25(3), 351–370.

Castelo, N., Bos, M. W., & Lehmann, D. R. (2019). Task-dependent algorithm aversion. *Journal of Marketing Research*, 56(5), 809–825.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340.

Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114–126.

Longoni, C., Bonezzi, A., & Morewedge, C. K. (2019). Resistance to medical artificial intelligence. *Journal of Consumer Research*, 46(4), 629–650.

Mayer, R. C., Davis, J. H., & Schoorman, F. D. (1995). An integrative model of organizational trust. *Academy of Management Review*, 20(3), 709–734.

McKnight, D. H., Choudhury, V., & Kacmar, C. (2002). Developing and validating trust measures for e-commerce. *Information Systems Research*, 13(3), 334–359.

Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1), 81–103.

Silge, J., & Robinson, D. (2017). *Text Mining with R: A Tidy Approach*. O’Reilly Media.

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.

Repository: <https://github.com/tyomachkaa/ai-acceptance-sharing-economy> · Online Content Analysis, WU Vienna, 2026.